

Agentic AI

Learners' Space 2026

Assignment 1

Topics: Understanding LLMs, Prompting and APIs

Submission Deadline: 16th June 11:59 pm

Hello! We hope you enjoyed the Week 1 resources on LLMs, Prompting, and APIs. This assignment will test your conceptual understanding and practical skills from those materials. Answer all the questions. Good luck!

Question 1: Read the following scenario carefully.

“You are building a chatbot for a hospital. The chatbot needs to remember a patient’s entire medical history (a 50 page document) while answering follow up questions. You are using an LLM with a 4,000 token context window.”

Why might this cause problems? What would happen if the medical history exceeds the context window? Suggest one practical approach to handle this limitation for the hospital chatbot.

Question 2: Write a Python script that uses the Anthropic API to do the following. (You may refer to the Anthropic API documentation linked in the resources).

Task: Build a simple ‘Study Buddy’ bot. The bot should:

1. Accept a topic as an input from the user (e.g. “photosynthesis”)
2. Send a request to the Claude API asking it to explain the topic in simple terms in under 100 words.
3. Print the response to the terminal.

Your script will be evaluated on:

- Correct use of the API (model, messages format, API key handling)
- Clean, readable code with comments

- The bot actually running and producing a sensible output
- Handling the case where the user provides an empty topic (basic input validation)

For submission: Submit your .py file and a screenshot of the output for topic = "neural networks".

Question 3: Below is a task: classify the sentiment of a product review as Positive, Negative, or Neutral.

“The delivery was on time but the product stopped working after two days. Disappointed.”

1. Write a zero-shot prompt for this task
2. Write a few-shot prompt for the same task that includes at least 2 example review-label pairs before asking the model to classify the given review.
3. Which prompt do you expect to perform better for an LLM, and why?

Question 4: Build a tool that will help IITB students plan their weekly study schedule. A student provides:

1. Their courses this semester (e.g. MA 110, CS 101, HS 101, etc.)
2. Number of hours available per day
3. Any upcoming exams or deadlines

Complete the following tasks:

1. Write a basic initial prompt for this task.
2. Identify two specific weaknesses or ambiguities in your initial prompt. Explain what could go wrong.
3. Write an improved prompt that fixes those weaknesses. Your improved prompt should specify: the output format (e.g. a table or a list), tone, and any constraints (like not scheduling more than 3 hours of one subject in a day).
4. Write a system prompt (1-3 sentences) you would give the model to set the right persona for this task.

Question 5: (Optional)

(Attempt this question only if you went through the Transformer resources).

In the sentence "The bank by the river was steep," explain in 2-3 sentences how the attention mechanism helps the model understand that 'bank' here refers to a riverbank and not a financial bank.

Submission Guidelines:

Submit a single pdf containing all your answers in the attached Google Form.

Name your pdf according to the following convention:

AgenticAI_Week1_<YourName>_<RollNo>

Google Form link (for assignment submission):

<https://forms.gle/ZeA6MXGhgLU2NifS7>

Resources for this assignment are listed on the Notion page. You are encouraged to re-watch or re-read any of them as needed. Reach out on the Whatsapp group or the MS Teams channel for any doubt.

All the best!